

Surname

Forename(s)

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS

PAPER 1

4030/1

NOVEMBER 2017 SESSION

2 hours 30 minutes

Candidates answer on the question paper.
Additional materials:
Geometrical instruments

Allow candidates 5 minutes to count pages before the examination.

This booklet should not be punched or stapled and pages should not be removed.

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces at the top of this page and your Centre number and Candidate number on the top right corner of every page of this paper. Check that all the pages are in the booklet and ask the invigilator for a replacement if there are duplicate or missing pages.

Answer **all** questions.

Write your answers in the spaces provided on the question paper using **black** or **blue** pens. If working is needed for any question, it must be shown in the space below that question.

Omission of essential working will result in loss of marks.

Decimal answers which are not exact should be given correct to three significant figures unless stated otherwise.

Mathematical tables, slide rules and calculators should not be brought into the examination room.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 30 printed pages.

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NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR CALCULATORS
MAY BE USED IN THIS PAPER.

1 Express 99,987

- (a) correct to one decimal place,
- (b) correct to one significant figure,
- (c) in standard form.

Answer:

(a)	_____	[1]
(b)	_____	[1]
(c)	_____	[1]

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- 2 From the following list of numbers write down the rational numbers;

3,1; $\sqrt{3}$; π ; 0,3; $\sqrt{169}$; $\frac{17}{19}$.

Answer:

[3]

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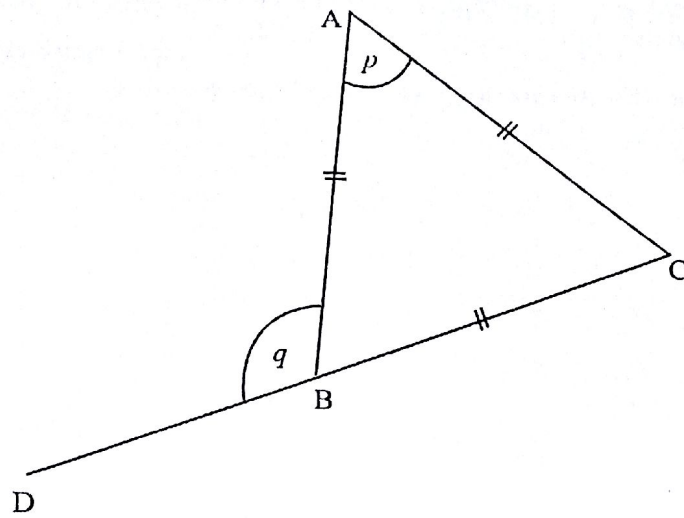
3 Simplify

(a) $(2 + x)^0$,

(b) $\frac{3x^{-2}}{3x}$.

5

4



- (a) Write down the special name given to triangle ABC.
- (b) Calculate the angles marked
- (i) p ,
- (ii) q .

Answer: (a) _____ [1]

(b) (i) $p =$ _____ [1]

(ii) $q =$ _____ [1]

- 5
- (a) Calculate the number of minutes between 1935 and midnight of the same day.
- (b) Write 75% as a common fraction in its lowest terms.

Answer: (a) _____ minutes

(b)

- 6 (a) Estimate the value of $\frac{7,63 \times 5,81 \times 9,3}{27,1 \times 31,2}$ by rounding off each number correct to one significant figure.
- (b) Find the Highest Common Factor (H.C.F) of 9 and 12.

Answer: (a) _____ [2]
(b) _____ [1]

7 Evaluate, giving your answer in its lowest terms,

(a) $\frac{4}{5} - \frac{9}{10}$,

(b) $\frac{7}{15} \times \frac{3}{8} \div \frac{7}{8}$.

Answer: (a) _____ [1]

(b) _____ [2]

9

- 8 (a) Express the ratio 3,5: 1 in the form, $a:b$ where a and b are integers.
- (b) Shuvai mixed hot water and cold water in the ratio 3,5: 1 to get warm water. If she ended up with 45 litres of warm water, calculate the amount of hot water she used.

Answer:

(a)

[1]

(b)

litres [2]

- 9 The mean of 9 numbers is 11,9. The mean of 8 of the numbers is 13.
Find the ninth number.

Answer: _____ [3]

- 10 Solve the equation $3x^2 - 5x - 2 = 0$.

Answer: $x =$ _____ or _____ [3]

- 11 (a) Find the square of $\frac{1}{4}$.
- (b) Calculate the rate at which \$500 earns \$320 simple interest in 4 years.

Answer: (a) _____ [1]

(b) _____ % [2]

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12 It is given that

the Universal set $\xi = \{x: 10 \leq x < 20, x \text{ is an integer}\}$,

subset $A = \{x: x \text{ is a prime number}\}$,

subset $B = \{x: x \text{ is a factor of 3}\}$ and

subset $C = \{x: x \text{ is an odd number}\}$.

- (a) Write down the largest element of subset A.
- (b) Find $n(B)$,
- (c) List elements of $A \cap C$.

Answer:

(a)

[1]

(b)

[1]

(c)

[1]

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13

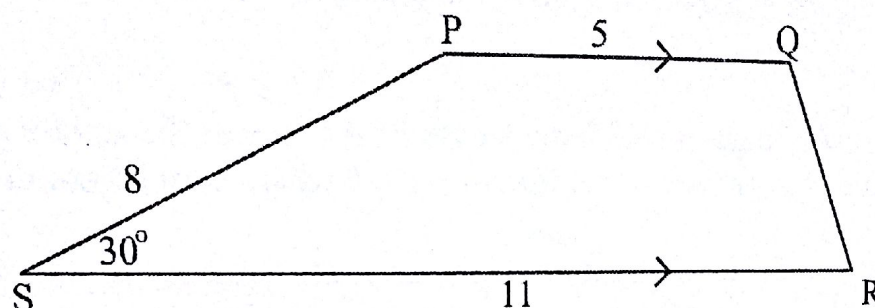
13 It is given that the length, l , of a side of a square is 7 cm, measured to the nearest centimetre.

- (a) Write the range in which the length of the side of the square must lie, giving your answer in the form $a \leq l < b$ where a and b are to be found.
- (b) Calculate the least possible perimeter of the square.

Answer: (a) _____ $\leq l <$ _____ [2]
 (b) _____ cm [1]

14

14



In the diagram, PQRS is a trapezium with PQ parallel to SR, $PQ = 5$ cm, $PS = 8$ cm, $SR = 11$ cm and $\hat{PSR} = 30^\circ$.

Calculate the area of the trapezium,

[Use as much of the information given below as is necessary].

[$\sin 30^\circ = 0,500$; $\cos 30^\circ = 0,866$; $\tan 30^\circ = 0,577$]

Answer:

cm²

[3]

- 15 (a) What is the number of lines of symmetry on a general parallelogram.
- (b) Find the exact value of $4mn$ when $m = 4 \times 10^4$ and $n = 5 \times 10^{-9}$, giving the answer as a decimal.

Answer: (a) _____ [1]

(b) _____ [2]

16

16 (a) Given that matrix $\mathbf{T} = \begin{pmatrix} 3 & -6 \\ 4 & 1 \end{pmatrix}$, find the determinant of matrix $\mathbf{T}$.

(b) If $\mathbf{n} = \begin{pmatrix} -3 \\ 9 \end{pmatrix}$ find $|\mathbf{n}|$, giving the answer in surd form.

Answer:

(a) _____ [1]

(b) $|\mathbf{n}| =$ _____ [2]

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- 17 (a) Given that $d = 2n - 1$, find d when $n = 8$.
- (b) On a map of scale 1: 50 000, a stream is 7 cm long.
Calculate the actual length of the stream, giving the answer in km.

Answer:

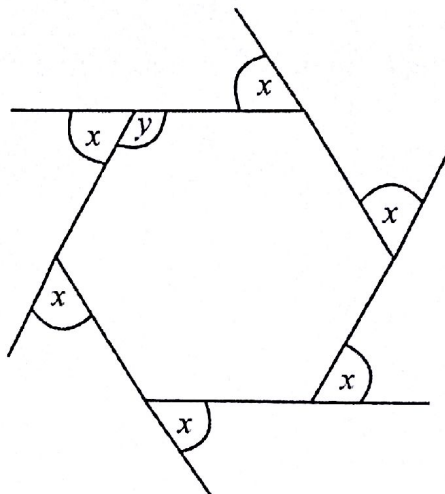
- (a) $d =$ _____ [1]
- (b) _____ km [2]

[Turn over]

18

18 (a) Write down the three-figure bearing equivalent to the direction North West.

(b)



The diagram shows a regular polygon with angles marked x and y .

Calculate the value of

(i) x ,

(ii) y .

Answer: (a) _____ [1]

(b) (i) $x =$ _____ [2]

(ii) $y =$ _____ [1]

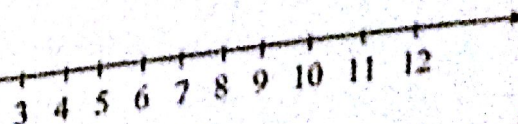
- 19 (a) Solve the inequality $3x + 1 < 28 \leq 5x - 2$.
- (b) Show the solution in part (a) on a number line.
- (c) If x is an odd integer write down its least value which satisfies the inequality in (a).

Answer:

(a)

[2]

(b)



[1]

(c)

$x \approx$ _____

[1]

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- 20 (a) Convert 1001_2 to a number in base 6.
- (b) Given that $111_n = 73_{10}$, find the value of n .

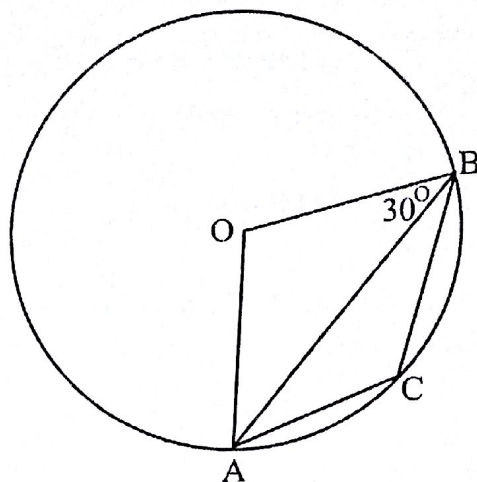
Answer:

(a) _____ [2]

(b) $n =$ _____ [2]

21

21



In the diagram, the points A, B and C are on the circumference of a circle centre O.
 $\angle OBA = 30^\circ$.

Calculate

- (a) $\angle AOB$,
(b) $\angle ACB$.

Answer: (a) $\angle AOB =$ _____ [2]
(b) $\angle ACB =$ _____ [2]

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- 22 (a) Matrix $R = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$. R represents a transformation that maps point $P(x, y)$ onto $P_1(3; 4)$.

Calculate the value of x and the value of y .

- (b) Given that $r = 27$, $s = \frac{1}{3}$ and $t = 2$, find the value of

(i) r^s ,

(ii) s^{-t} .

Answer:

- (a) $x =$ _____
 $y =$ _____ [2]
- (b) (i) _____ [1]
(ii) _____ [2]

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- 23 (a) Factorise completely the expression $4x^2 - y^2$.
- (b) Given that $2x + y = 10$, solve the equations:

$$4x^2 - y^2 = 40$$

$$2x + y = 10$$

Answer:

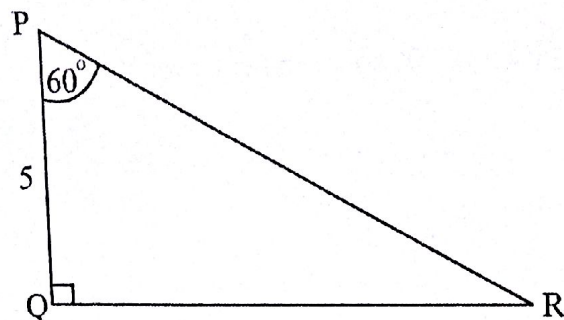
(a) _____ [2]

(b) $x =$ _____

$y =$ _____ [3]

24

24



In the diagram $\angle PQR = 90^\circ$, $PQ = 5$ cm and $\angle QPR = 60^\circ$.

- (a) Calculate PR .

[Use as much of the information given below as is necessary]

$[\sin 60^\circ = 0,866; \cos 60^\circ = 0,500; \tan 60^\circ = 1,732]$

Answer: (a) $PR =$ _____ cm [2]

24 (b) Solve the equation

(i) $0,2 n = 5,$

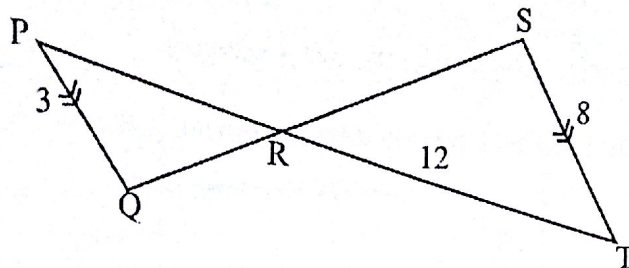
(ii) $\frac{1}{3} m + 2 = 8.$

Answer: (b) (i) $n = \underline{\hspace{2cm}}$ [1]

(ii) $m = \underline{\hspace{2cm}}$ [2]

28

26



The diagram shows two triangles PQR and RST such that PQ is parallel to ST, $PQ = 3$ cm, $ST = 8$ cm and $RT = 12$ cm. PRT and QRS are straight lines.

- (a) Name the triangle similar to triangle PQR.
(b) Calculate PR.

Answer: (a) _____ [1]

(b) $PR =$ _____ [2]

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- 26 (c) Triangle RQP is mapped onto triangle RST by a transformation A.
Describe fully transformation A.

Answer:

(c)

[3]

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- 27 When a rod of length l cm is heated to a temperature of $t^{\circ}\text{C}$, it expands. Its new length, x cm, is given by the formula $x = l + alt$.

- (a) Make a the subject of the formula.
- (b) (i) Find x when $l = 2$ cm, $a = 0,01$ and $t = 50^{\circ}\text{C}$.
- (ii) Find a when $x = 3,1$ cm, $l = 3$ cm and $t = 40^{\circ}\text{C}$.

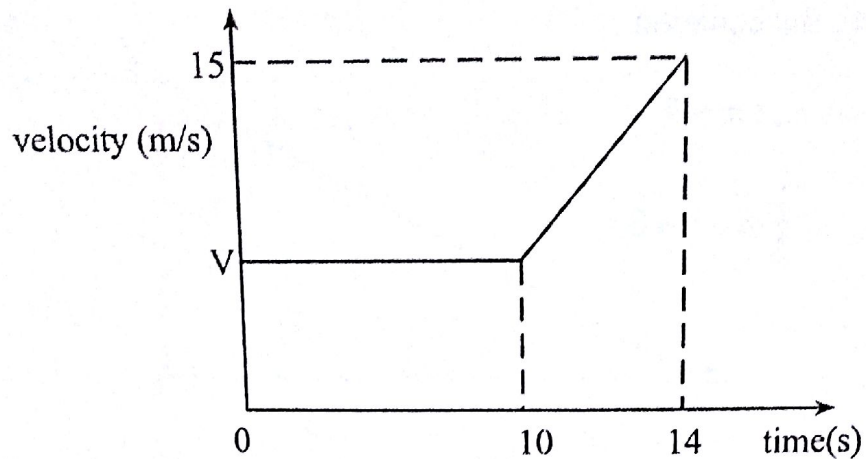
Answer: (a) $a =$ _____ cm [2]

(b) (i) $x =$ _____ [2]

(ii) $a =$ _____ [2]

26

25



The graph shows part of a velocity-time graph of a moving object. The object travels at a constant velocity of V m/s for 10 seconds. It then accelerates uniformly until it reaches a velocity of 15 m/s in 4 seconds.

- (a) The distance travelled during the first 10 seconds is 50 m.

Calculate V .

Answer: (a) _____ m/s [2]

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- 25 (b) Find the
- (i) acceleration during the last 4 seconds,
 - (ii) total distance travelled by the object in the 14 seconds.

Answer: (b) (i) _____ m/s^2 [2]
(ii) _____ m [2]